AC9M6ST03 • Year 6 Mathematics

Planning and Conducting Statistical Investigations

Pose questions, collect representative data and communicate justified findings

We are learning to…

Students define a statistical question and population, choose sampling and measurement methods, clean data, select displays, compare distributions and report findings with limitations.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

plan and conduct statistical investigations by posing questions, collecting or accessing data, selecting and creating displays and interpreting results in context

Use a complete investigation cycle

1. pose statistical question
2. define population/variables
3. plan sample/method
4. collect and clean
5. represent
6. analyse
7. conclude
8. evaluate and communicate

The question determines the data and display. A strong conclusion is proportionate to the sample and method.

Plan sampling and data quality

decision	quality question
population	**Who or what is the conclusion about?**
sample	**How can selection be fair and practical?**
variable	**Categorical, discrete or continuous?**
measurement	Unit, precision and consistency?
cleaning	Duplicates, missing values, categories?
report	Finding, uncertainty and limitation?

Digital tools support collection and analysis, but they cannot repair a leading question or biased sample automatically.

Build and annotate the model

Represent planning and conducting statistical investigations and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Question has one fixed answer:** Statistical questions expect variability.
- **Convenience sample generalised broadly:** State selection limitations.
- **Data cleaned by changing inconvenient values:** Correct only documented errors.
- **Conclusion written before analysis:** Let evidence guide it.

Quick check

- Write a statistical question.
- Define population/sample.
- Choose variable type.
- Plan a display.
- State a limitation.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.